EARLY LEARNER CURRICULUM AND ACHIEVEMENT RECORD (ELCAR): A CABAS® Developmental Inventory

Manual and Curriculum Guide

THIS IS THE 7TH REVISION OF WHAT WAS FORMERLY TITLED

CABAS® International Curriculum and Inventory of Repertoires for Children from Preschool through Kindergarten (C-PIRK)

Administration Manual and Curriculum Guide- Sixth Edition

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This version takes precedence over all prior versions by R. Douglas Greer

A portion of the proceeds from the sale of the ELCAR will go to the Fred S. Keller School. The Keller School is named in honor of Dr. Fred S. Keller, the founder of modern reinforcement theory and a distinguished educator who led the field in the application of the science of behavior to reform educational practice. The Keller School was founded by Dr. R. Douglas Greer in 1986 as a CABAS® model preschool. Today, The Fred S. Keller School has three locations in Rockland and Westchester Counties in New York and serves approximately 400 students, ages 16 months to 5 years, with and without language and developmental disabilities. The Keller School campuses serve as Research and Demonstration sites for Columbia University Teachers College programs in teaching as applied behavior analysis.

Acknowledgements

We would like to thank the following individuals for their assistance in early drafts of the ELCAR Achievement Record: Jennifer Lee, Erika Byers, Elizabeth Snell, and Jo Ann Delgado

Foreword

By R. Douglas Greer

I began writing the early drafts of what became the PIRK (Preschool Inventory of Repertoires for Kindergarten) in 1986, when I was asked to start a preschool that eventually became the Fred S. Keller School. Nan McCorkle provided the first version of the motor skills section for that first draft. Over the years that section was almost entirely rewritten, but Nan made the first steps possible, so to speak. Many individuals have contributed to what became the C-PIRK (CABAS® International Curriculum and Inventory of Repertoires for Children from Preschool Through Kindergarten), suggesting new objectives as well as the elimination of others. Still others assisted with ideas for formatting to improve the ease of use. Claire Cahill, Nirvana Pistoljevic, and Jeanne Speckman provided many of these suggestions. Our colleagues in our international schools also helped make the C-PIRK more international and multicultural. Most recently Robin Nuzzolo, Jeanne Speckman, Lin Du, and Jennifer Longano have been active in developing videos for calibration and training purposes. I thank Jessica Dudek for proofing and editing content areas. Without the latter efforts, I would have been reluctant to make the C-PIRK available to the professionals outside of the CABAS® system. Of course, I am appreciative of the thousands of children who have received the various versions of this curriculum and assessment over three decades for teaching us to make it better. I also thank the hundreds of our CABAS® teachers and teacher assistants, doctoral and masters students who showed that direct and reliable measurement of children's mastery of critical instructional objectives can be monitored reliably day in and day out for decades.

This edition is the result of the work of a committee of Supervisors and Consultants of the Fred S. Keller School in 2018 and 2019. They revised, expanded. and improved this curriculum and repertoire inventory. This resulted in a change in title and the addition of coauthors. The new title was selected because it was thought to be more representative of the contents and more accessible to a broader audience. It is now the "Early Learner Curriculum and Achievement Record (ELCAR): A CABAS® Developmental Inventory.

Methinks, in the best of all possible worlds, the actual measurement of educational outcomes will be the platinum standard of assessment. Rather than a score indicating where a measure constitutes how accurate the student is among a general population, repertoire assessments indicate what the person has mastered. Standardized achievement tests are valuable indications of the distribution of learning in particular populations and provide useful information. But they assume that the function of education is to provide exposure to the same degree across all students with some learning more than others. From the perspective of *a strategic science of teaching, individuals are instructed until the demonstrate mastery and it is the responsibility of the* instructional team, within the constraints of what the science provides, to ensure the student's mastery. Because of individual differences in verbal development, and the constraints of time, some children will master more objectives than others. The term *mastery* is defined as given the relevant motivating conditions and context the student will accurately emit the behavior. Other repertoires refer to the individual's community of reinforcers that includes the things and social stimuli the student has come to be interested in as a result of teaching procedures.

I trust that the ELCAR is a step in that direction.

About the ELCAR

The CABAS® ELCAR constitutes the curricular objectives for teaching the repertoires that are necessary for a child to excel in kindergarten and the primary grades, including the repertoires associated with the physical, emotional, social, academic, language development and enrichment.

In addition to being a curriculum, the ELCAR is also a curriculum-based inventory and criterion referenced assessment. It is a direct measure of what the child has learned to do and the child's rate of progress in mastering the curriculum. Of course, it is also a measure of how well the instructional professional team has taught the child. This is a useful tool, but like all tools, the user must know how, where, and when to use it and for what purpose. For that reason, we have developed instructional videos that provide professionals with sequential instruction leading to the reliable and valid use of the curriculum and assessment.

The ELCAR provides a list or inventory of what the child knows and what the child needs to learn. The ELCAR is a criterion-referenced assessment of the child's mastery of curriculum. A long-term-objective in the ELCAR is a repertoire. The term *repertoire* includes not only the behaviors but also the antecedents, or what comes before the behavior, and consequences of behaviors, or the outcome of the behavior, within specific contexts. Together, the behavior and the contexts for the behavior are operants and it is operants not just behaviors that are learned. The contexts include setting events such as motivational conditions within which the student's operants occur. One cannot determine what has been learned or not learned without knowing all aspects of the behavior in context. Moreover, what reinforces the behavior is as important, if not more so, than the actual behavior that is learned. Thus, these are critical components of what the ELCAR assesses. Once an objective is marked as having been obtained, then the child has mastered that objective under the conditions specified in the assessment.

The ELCAR is designed for instructional use and to provide professionals and parents with an in-depth assessment that can be used to further the child's education in a systematic fashion. This type of curriculum-based assessment of children's learning should then drive the work of the professionals and parents. The ELCAR should only be implemented by those who have demonstrated competencies associated with its theoretical and practical applications. These competencies include achievement of mastery criterion on the written ELCAR quiz and calibration to 100% accuracy on the ELCAR training videos.

The ELCAR is not a standardized test. However, if well-trained professionals use it reliably, it provides a more complete and accurate measure of an individual child's repertoires or knowledge than does a standardized test score. This is because the ELCAR is a direct measure of what the child has learned, whereas a standardized test score is an estimate. The standardized score is more useful in assessing an entire classroom or a particular school's standing relative to the national or state norms for which the standardized test was designed. Both measures are important, but the actual instructional outcomes and needs for the individual child are more valid when the measurement is a criterion referenced one.

Included in the ELCAR are also directives to stop assessment at certain points, as well as to temporarily place a hold on instructional objectives so that we may assess whether a specific developmental cusp or learning capability is present or absent. This enables users to determine whether to implement one of the verbal behavior protocols which will be referenced in several locations of the ELCAR. Further discussion and description of cusps, capabilities and protocols appears later in this section.

The ELCAR is an appropriate and validated assessment tool for determining treatment goals and objectives within the context of designing clinically based ABA services for individuals with Autism and related disorders. In addition the ELCAR provides a scientifically sound framework for indicating when these services may be reduced and when clients may transition to less intensive services.

Because the main barriers to children learning involve barriers that prevent children from accessing educational materials, an appropriate and effective curriculum and achievement record must focus on the elimination of these barriers. Many of the behaviors that maintain them are not traditional problem behaviors but are rather missed or underdeveloped milestones or cusps. The ELCAR, paired with Verbal Behavior Protocols, allows us to address the core deficits of Autism and remove barriers to access and learning by targeting these cusps.

The ELCAR was developed at sites that service children with varying degrees of developmental delays and Autism, as well as neurotypical children. Several portions of the Inventory have been designed to target behaviors specific to Autism Spectrum Disorder. The reinforcer/preference assessment section focuses on stimuli that children with Autism tend to approach or avoid. The observing response section, which includes such items as "joint attention" and "orients towards the face of a person", focuses on crucial behaviors that are commonly missing in individuals with Autism, but are necessary for setting the proper foundations for verbal behavior. By targeting missing behavioral cusps and capabilities through scientifically tested verbal behavior protocols, the ELCAR is an effective tool in removing educational barriers.

The ELCAR should be implemented according to the procedures outlined in *Designing Teaching Strategies* (Greer, 2002) and *Verbal Behavior Analysis* (Greer & Ross, 2008). The user should also be familiar with the unit of measure we call the learn unit (Greer, 2002). While the curriculum provides a list of the repertoires in each domain, the procedures for teaching the curriculum (deriving the sub-objectives) are the result of decades of inductive research leading to a strategic science of teaching for all kinds of learners. These procedures, as well as the philosophical rationale and research base, are described in detail in the books *Designing Teaching Strategies: An Applied Behavior Analysis Systems Approach* (Greer, 2002) and *Verbal Behavior Analysis* (Greer & Ross, 2008). Further, throughout the ELCAR we refer the teacher to the Verbal Behavior Developmental interventions (protocols) when instruction is not successful using standard best practices from the sciences of learning, teaching, and behavior management. It is not possible to for students to learn certain repertoires in this, and other curricula, if critical developmental cusps and learning capabilities are missing. The *Verbal Behavior Analysis* book and related instructional videos will prove useful. It is important that instruction related to the missing developmental cusps not be continued when these critical developmental cusps and learning capabilities are not present.

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I. COMPONENTS OF THE ELCAR

A. Screenings

1. Preference/ Reinforcer Screening

It is crucial to identify which environmental stimuli function as reinforcers and potentially aversive stimuli for early learners. The sooner professionals gather this information the sooner they can establish the correct motivational conditions for learning. The preference/ reinforcer screening provides opportunities to observe students' responses to different common classroom items and activities. These responses are coded as "avoid," "neutral," "potential," and "no response" and the data should be used to guide professionals' choices of materials and activities when providing instruction.

2. Observing Responses Screening

Observing responses are foundational to the development of early language (Keohane, Pereira-Delgado, & Greer, 2009). Observing responses are operants, as they are selected out by the environmental stimuli in the moment. These include operants such as looking, listening, tasting, smelling and touching. The observing response screening outlined in the ELCAR is designed to 1) determine which types of stimuli in the environment set the occasion for observing responses for a specific learner, and 2) serve as a possible indicator that a verbal cusp or capability is either present or absent. The observing response screening is **not** a full assessment of whether cusps are present, but they may direct the professional to conduct a full assessment (found in the protocol packages). It is important to note that these observing responses should not be directly taught; instead once stimuli acquire reinforcing properties these stimuli will then select out the responses.

3. Verbal Operant Screening

When conducting the verbal operant screening, we observe students across several settings and activities and record their emission of verbal operants. The screening is conducted across three activities and school settings (toy area/ non-instructional, lunch time, and group setting with teacher directed activities). The lexical and non-lexical utterances are divided into the following categories: tact, mand, echoic, intraverbal, textual response, and self-talk during fantasy play. The function of the verbal operant screening is to assess which verbal operants students emit under which conditions, and how they function for the student (the purpose of categorizing). A Verbal Operant Data Collection Form is provided in the Materials Packet.

4. Instructional Readiness Screening/ Sequence (IRS)

Educational professionals must ensure that that the correct environmental stimuli are selecting out student responses during assessments. This, along with determining the correct motivational variables, is crucial to a reliable assessment of students' repertoires. We use two- and three-dimensional stimuli, physical gestures and vocal directions during the instructional readiness screening. The main objective of the IRS is that the instructor's presence establishes the conditions necessary for student responding. Professionals use vocal directions paired with gestural cues across a variety of functional

activities. If the student does not respond with 50% or greater accuracy, teach these responses using learn units until the student responds with at least 50% accuracy. This is a good indicator that a reliable ELCAR assessment may be conducted.

B. Achievement Record

1. When to conduct the initial assessment

When conducting the ELCAR, you must always ensure that the child's lack of correct responses is not a function of the fact that the child has not adjusted or habituated to the school and the teacher. Thus, begin the ELCAR Achievement Record only after the child has adjusted to the classroom and the teacher. We measure this by conducting the Preference/Reinforcer Screening and Instructional Readiness Screening. Once you have identified potential or actual reinforcers, and the student is responding with at least 50% accuracy to the Instructional Readiness Screening (recommended criterion), proceed with the initial assessment. Given the proper motivating conditions, the ELCAR Achievement Record will be a reliable predictor of what the child does and does not know at any given point in time. See the Materials Packet for data collection forms with instructions for some of the ELCAR items.

2. Why provide consequences during the assessment?

The majority of repertoires in the ELCAR are measured using Learn Units. Presenting learn units ensures that if the child does have the repertoire, the child's responses will not be placed into extinction due to a lack of reinforcement. Correct responses are reinforced under the specified reinforcement condition. For example, if the objective specifies generalized reinforcement, then the behavior must be performed under approval or token economy conditions. The assessment of the type of reinforcer that is or is not effective is just as important, if not more so, than the emission of the behavior. The target reinforcing consequences are listed in the ELCAR Achievement Record each of the response categories. Before administering the assessment, please check the type of reinforcement condition specified under each section. Please refer to the Learn Unit training videos.

The categories of consequences that reinforce the behavior in particular contexts include:

Prosthetic Reinforcers (PR)- emitting the behavior under prosthetic reinforcement conditions (i.e. edibles, toys, activities, or other consequences that are not the natural result of behavior).

Generalized reinforcers (GR)- emitting the behavior under teacher approval and token reinforcement conditions. In many cases the natural reinforcement is the goal. However, in the case of many academic responses, such as reading, the eventual natural reinforcement takes many months and even years to reinforce the behavior. Thus, we accept generalized reinforcers as appropriate reinforcers in those areas indicated in the ELCAR.

Natural Reinforcers (NR)- emitting the behavior under natural reinforcement conditions that are the naturally occurring results of operants (e.g., requesting something as a "mand" results in the item being delivered). When an operant is truly part of one's repertoire, the natural effect of the behavior is the reinforcement. For repertoires associated with the community of reinforcers, such as looking at books or playing with toys, the target consequence is to be the stimuli controlling the responding or observing behavior, rather than teacher praise or tokens.

Reinforcement	**Examples**	**Abbreviation**
Prosthetic	Edibles, Toys, or Activities	PR
Generalized	Praise or Tokens	GR
Natural	Natural Consequences	NR

Corrections are delivered for incorrect or no responses. However, because the assessment typically does not allow for more than one opportunity to respond to each target stimulus, the effects of the correction procedure on future responses will not be immediately known. See the ELCAR training video and accompanying materials for thorough coverage of the learn unit and implementation procedures.

You may find that, in cases where opportunities to emit the same response topography (e.g., match-to-sample responses) or where the antecedent is the same (e.g., motor imitation), the child begins to learn from the learn unit presentations during the assessment. That is, after initial incorrect or no responses, the child emits a correct response that is reinforced, followed by more correct responses. This may indicate an initial lack of stimulus control and not that the child did not know the response and is fortuitous because it suggests that the repertoire may be quickly mastered or, in fact, may be present. If you observe learning occurring, continue the assessment until criterion has been achieved, that is, until the child emits correct responses for each operant or achieves the pre-specified criterion (e.g., 5 consecutive correct match-to-sample responses). There are many sections of the ELCAR where learn units are not used as part of the assessment. These sections are indicated when the reinforcement is under "natural reinforcement (NR) contingencies (e.g., Community of Reinforcers).

3. Coding the Achievement Record

Item Number			Criteria	Date(s) Identified as Absent	Date Identified as Present
V	**Self-Management**				
A	**School Self-Sufficiency Repertoires**				
	All repertoires below should be under Natural Reinforcement (NR) contingencies				
1	Stays in play area for 10 minute	In order to mark this item achieved, the student must stay in the classroom toy area for the entire 10-minute session without teacher prompting or reinforcement across two consecutive sessions.	2/2	- 9/15/19 RDG	+ 10/1/19 RDG
2	Remains seated at the table for individual instruction during a 5-minute observation	This probe should be conducted during instruction as appropriate for the student, with Learn Unit instruction.		- 9/15/19 RDG	
3	Remains seated at the table while the teacher delivers instruction to another student during a 5-minute observation	The student should be given an activity that can be completed independently.		NA	

Assessment of repertoires

There are three components to the ELCAR code:

1) +, -, NA 2) Date 3) Initials

➢ The criterion for each repertoire is specified

➢ When a student is assessed and the repertoire is present, record a "plus"**,** the date, and your initials in **black ink** in the "Present" column **(+10/1/19 RDG**).

➢ When a student is assessed and the repertoire is absent, record a "minus", the date, and your initials in **black ink** in the "Absent" column (**-9/15/19 RDG**).

➢ If the repertoire was not assessed, or not currently possible to be assessed due to missing prerequisites, **record an "NA" (Not Assessed) in pencil.** You do not need to include the date or initials when recording "NA". The notation in pencil may be erased and replaced when the repertoire can be assessed.

4. Displaying results

The ELCAR Achievement Record includes a bar graph for displaying the number of mastered repertoires. The curricular domains are divided up by category. You will fill in the number of repertoires identified as present, so that the results are displayed cumulatively.

II. EDUCATIONAL SEQUENCE

The ELCAR Achievement Record is designed to function as both an assessment and a curriculum. The initial inventory establishes the areas in which the student requires instruction or intervention, and the curriculum guides the sequence of educational objectives towards long term functional goals.

A. Curriculum

1. What does the curriculum include?

We have attempted to create a comprehensive list of objectives and standards for early learners. The curricular objectives in the ELCAR are derived from the educational standards of several countries. These standards have been converted to functional repertoires. This means that each standard and ELCAR repertoire incorporates a structural goal, as is the usual case for educational standards, and a functional goal. This distinguishes the ELCAR from other curricula because the items mastered will result in functional repertoires across a variety of contexts. In addition to the international standards, the CABAS® standards that cover self-management and the child's community of reinforcers are included.

2. How is the curriculum organized?

The curriculum and inventory are arranged according to verbal behavior development. In most cases, repertoires in each domain are listed from least to most difficult. However, do not presume that more advanced repertoires require all of the prerequisites.

3. How to set up the correct instructional sequence

For many items, the ELCAR inventory includes a sample list of responses that are representative of a repertoire. Hence, the assessment criterion may be 5/5 representing that the student has 5 target operants in repertoire for a specific item. However, when we are establishing curricular objectives to target an ELCAR item, a session of instruction may include 20 learn units across 5 operants presented 4 times each, and the criterion for mastering that objective may be preset at 90% across 2 consecutive sessions. Consider the following examples:

ELCAR Inventory Code:
X. Academic Literacy - B. General Knowledge- 50. Clothing Items on self as a listener

Mark this as present if there are 10 correct out of 10 presented clothing items. If this does not occur, a curricular objective may be established that looks like this:
20 learn units, 5 different clothing items presented 4 times each with a criterion for mastery of 2 consecutive sessions of 90%.

X. Academic Literacy- A. Matching- 1. Visually matches novel stimuli- a. colors

Mark this as present if there are 10 correct out of 10 presented match-to-sample responses across 10 different colored stimuli. If this does not occur, a curricular objective may be established that looks like this:
20 learn units, multiple pairs of items across 5 different colors presented 4 times each with a criterion for mastery of 2 consecutive sessions of 90%.

In these examples, keep in mind that criterion for mastery of an instructional objective IS NOT THE SAME as the criterion for the presence of the repertoire as listed in the ELCAR.

4. Materials/stimuli

When teaching between and within auditory, vocal, visual, olfactory, gustatory, and kinesthetic stimuli, it is critical to *rotate the irrelevant aspects* of the discriminated operant, while including the *essential element*. For example, when teaching "redness" the stimuli should include red in different objects, shapes, sizes, and 2- and 3-dimensions. Alternately, if you are teaching shapes the presentation of shapes should vary colors, sizes, context, and texture. For example, when teaching listener and speaker responses to sight words, different types of fonts and size and color of fonts might be used to teach the abstraction across speaker and listener responding.

Always present a minimum of two comparison stimuli along with the target stimulus when assessing or teaching match and point responses. Exceptions would be assessing or teaching differentiations between 2 stimuli, such as 1) differentiations between opposites (e.g., big and little) or 2) differentiations between stimuli based on how they compare to one another (e.g., more and less). When assessing or teaching match and point responses, always 1) say the word for the stimulus to be matched and 2) rotate the negative exemplars or comparison stimuli and the positioning of stimuli for each learn unit (be aware of patterns of responding such as right or left preferences).

5. Student data analysis

One of the most important decisions a teacher will have to make is when to continue instruction and when to stop and conduct an analysis of a potential learning problem. We refer to this as the Decision Analysis Protocol (see Appendix B). The decision protocol provides an algorithm for determining the source of the problem within the learn unit context. The first step is a graphic analysis of the data, which ensures that ineffective instruction does not continue indefinitely. If a decision is made to stop and analyze the student's data, the algorithm provides a series of strategic questions for the teacher to ask related to whether learn units are in place, setting events or the student's instructional history are interfering, or whether the student is missing a verbal behavior developmental cusp. Please refer to the Decision Analysis Protocol in Appendix B and CABAS® Decision Analysis Supervisor Script in Appendix C.

B. Pedagogy

Pedagogy is the study of the teaching operations. Research findings related to the study of effective pedagogy include not only the best operations to use to teach (i.e., direct learn units), but now include how particular verbal developmental cusps that children demonstrate predict how they can be taught best, or how they can learn even if the instructional presentations are not optimal. Thus, in a strategic science of teaching, it is critical to consider not only the repertoires that need to be taught, but how the child can best be taught according to the verbal developmental cusps present in his/her repertoire.

1. How verbal development informs pedagogy

> Rosales-Ruiz and Baer (1996) identified a behavioral cusp as:
> "a change that is (1) often difficult, tedious, subtle or otherwise problematic to accomplish, yet (2) if not made means little or no further development is possible in its realm (and perhaps in several realms); but (3) once it is made, a significant set of subsequent developments suddenly becomes easy or otherwise highly probable which (4) brings the developing organism into contact with other cusps crucial to further, more complex, or more refined development…" (p.166).

A developmental cusp is the result of a newly conditioned reinforcer(s). Several preverbal foundational cusps have been identified that are necessary for learning through learn unit instruction. These early cusps are observing responses that make subsequent learning possible. A child cannot respond to stimuli if he/she does not observe them. These include 2- and 3-dimensional stimuli as well as adult faces and voices. Conditioning those stimuli as reinforcers will result in those stimuli selecting out observing responses. Other cusps have been identified that are necessary for the child to become a true listener and speaker, such as discriminating the auditory properties of speech or echoing the speech of others (hear/say). Cusps indicating the joining of the initially separate listener and speaker repertories include verbal operants between or within individuals, such as self-talk or conversational units. Social learning cusps include those that involve indirect contact with observed contingencies. Finally, when print joins cusps, the teacher needs to establish conditioned reinforcement for reading and writing. A complete list of cusps can be found in Appendix A. A visual display for indicating the cusps that a child has in repertoire can be found in the Materials Packet.

Throughout the ELCAR achievement record, you will find the following symbol,  , to indicate the possibility of a missing cusp and a reference for the corresponding protocol. A verbal developmental protocol is an intervention that is put in place when it has been determined that a particular cusp is missing for a student. The function of a protocol is to establish the developmental cusp. This then allows the child to be taught objectives that he could not be taught when he or she was missing the cusp.

A verbal developmental protocol is different from the curricular objectives contained in the achievement record. A curriculum is designed to teach new repertoires

whereas a protocol is designed to establish a developmental cusp. Often the cusp is the result of the student learning new reinforcers. This may occur while teaching new repertoires (increasing tacts may result in learning a new reinforcer which is social praise and approval), but in many cases requires a specific intervention, which we call a protocol. For example, after adult voices have been conditioned as reinforcers for observing, the child can learn listener responses (hear/do) at a faster rate than before voices were conditioned as reinforcers.

We determine the need for protocols by (1) analyzing data from observing response probes, (2) analyzing students' curricular progress (slow rates of learning or lack of learning for certain objectives may suggest a missing cusp), and (3) conducting full length probes that test for the presence or absence of responses that usually accompany the cusp (e.g., visual tracking of objects usually indicates that three-dimensional stimuli function as reinforcers for visual observing responses.)

Examples of preverbal foundational cusps include conditioned reinforcement for observing adult faces and voices and 2- and 3-dimensional stimuli. Examples of listener and speaker cusps include listener literacy (hear-do) and mand and tact operants. Examples of verbal cusps indicating the joining of the speaker and listener include bidirectional operants between and within people and bidirectional naming (BiN). Cusps enable learners to come in contact with new contingencies and often are the result of newly learned or conditioned reinforcers. A cusp that is also a developmental capability allows an individual to learn in ways he/she could not before. Cusps that meet the definition of capabilities are those associated with Generalized Imitation, Observational Learning, and Naming. See Appendix A for a complete list of cusps.

We distinguish between a cusp and a repertoire. A repertoire is a group of responses, along with their antecedents and consequences, that belong to the same category. For example, tacting colors and answering a variety of "Wh" questions both constitute repertoires. The repertoires in the ELCAR can be treated as long-term objectives when they are identified as absent. The acquisition of a repertoire does not imply that reinforcers have changed or that a student can learn in new ways; however, the acquisition of a cusp can lead to an accelerated rate of learning of new repertoires.

In addition to considering the child's level of verbal development, we must use the correct type of instruction when addressing missing repertoires. The term for instructional presentations that meets the criteria for best practices is the *learn unit*. There are three broad categories of learn units that apply to the teaching of the repertoires in the ELCAR and in developmental assessments and interventions: *Direct Learn Units* (DLU), *Model Demonstration Learn Units* (MDL), and *Instructional Demonstration Learn Units* (IDLUs). Some objectives (e.g., the category of Community of Reinforcers) do not involve learn units.

2. Types of learn units to use relative to the student's verbal development

The determination of how a child should be taught (e.g., whether the child should be taught as part of a group or in one-to-one instruction) is related to the particular cusps that are also new learning capabilities that the child demonstrates. Thus, the presence or absence of particular cusps suggests the best type of learn units to use. For example, students with BiN can not only learn from group instruction learn units (e.g., response boards, choral responding) but may, in fact, learn faster. UniN and BiN allow students to learn from observing other students receiving instruction prior to the opportunity to respond (i.e., learn from antecedents). Students can also learn from observation of the consequences received by others if they observe *correct responses and incorrect responses*. However, when students observe only reinforcement for correct responses, they are less likely to learn from observing the responses received by other students. When students demonstrate BiN and observation of consequences received by others, they profit most from group learn units because they learn from observation of teacher instructions with examples even before they receive consequences, and they also learn from observing others receiving post response instructional corrections and reinforcement. Conversely, students who lack BiN and observational learning will not learn at all in group instructional settings.

Direct Learn Units (DLU) consist of an antecedent presented by a teacher or device, the student's response, and either a reinforcer for a correct response or a correction procedure followed by an unconsequated opportunity to emit an independent response if the original response was incorrect.

Model Learn Units (MLU) consist of the teacher modeling (see-do) the correct response. The teacher's model is the antecedent for the student to emit an imitative response. Correct responses are reinforced; incorrect responses involve the teacher again modeling the correct response. MLUs should be used under appropriate motivating conditions (e.g., morning routine, handwashing) such that when the model is removed the student will continue to emit the behavior under relevant conditions.

Instructional Demonstration Learn Units (IDLU) consist of the teacher providing an antecedent and demonstrating the correct response for two different exemplars of the objective. After two IDLUs, DLUs are presented. IDLUs should be used primarily to teach concepts or abstractions (e.g., intraverbals, autoclitics, categories).

Examples of Learn Units

DLU	**MLU**	**IDLU**
Example: Student will point to body parts. Teacher says: "Point to your head." Student: Student responds by pointing to his/her head.	*Example: Teach a student handwashing routine.* Teacher models handwashing routines by modeling all of the steps. Student observes the teacher completing all steps of handwashing.	*Example*: *Student will tact cat, dog, bear, fox, bird.* Teacher holds up a picture of a cat and says: "This is a cat." Teacher holds up picture of a dog and says: "This is a dog."

Teacher: Provide reinforcement if student points to head. Provide correction if student does not point to head.	Teacher models the correct response. If the student emits an incorrect response, student receives a correction. If student emits a correct response, student receives reinforcement.	Teacher holds up a picture of a bear and says: "This is a bear." Teacher holds up picture of a fox and says: "This is a fox." Teacher holds up a picture of a bird and says: "This is a bird." "Let's look at these pictures again." After two IDLUs per operant, student receives independent opportunities to respond (direct learn units).

Use Direct Learn Units (DLUs) when the child demonstrates:
(a) Instructional Readiness (ELCAR Inventory Code: **IV. Instructional Readiness Screening- A. 1)**, which is not a cusp but the prerequisite for doing anything with children. This means that the student is under the instructional control of the teacher. The teacher has identified and correctly uses operations that reinforce the child's responses and attention. Moreover, the face and voice of the teacher and the instructional stimuli themselves select out the child's attention, which is a preverbal foundational developmental cusp that should be present or established.

Use Model Learn Units (MLUs, i.e., show child how to do the correct response) for certain curricular objectives as antecedents to Direct Learn Units when the child demonstrates:
(a) A history of benefitting from DLUs and demonstrates or acquires
(b) Generalized Motor Imitation. This should be done for objectives such as those that involve construction or assembly, riding bikes, folding clothes, but not the discrimination of, or manipulation of, 2 dimensional stimuli (e.g., writing, drawing, matching).
(c) Generalized imitation and emulation, not just generalized *motor* imitation. if the child demonstrates fine motor imitation, facial imitation, and echoic responding they will likely benefit from models. In this case, prior to a lesson, demonstrate accurate responding to two exemplars prior to beginning DLUs.

Use Instructional Demonstration Learn Units (IDLUs) only if the child demonstrates UniN or BiN.

Use Group Instruction with IDLU learn units when the child demonstrates:
(a) Observational Performance (responds with group by observing reinforcers for correct responses and corrected responses.
(b) Observational Learning - operants or reinforcers

ELCAR Quiz

Respond to the following questions. You may refer to the ELCAR Manual, Achievement Record, and Materials Packet. Mastery criterion is 90% accuracy.

1) List the curricular domains in the ELCAR (Hint: headings use Roman numerals) and provide an example of two repertoires from each.

2) There are four Screenings included in the ELCAR. List and briefly explain the function of each.

3) The ELCAR is an assessment tool, but also a curriculum. Explain what this means.

4) Explain the benefits of a criterion-referenced assessment.

5) You must always ensure that a child's lack of correct responses is not a function of the fact that the child has not habituated to the school and the teacher. Explain what this means. What two screenings determine whether a child is ready for the assessment?

6) Define and explain how these terms are different in the ELCAR:

    a) Repertoire:

    b) Behavior:

    c) Cusp:

7) Why does the ELCAR specify the type of reinforcement for each repertoire? Describe and give an example of each type of reinforcement:

    a) PR
    b) GR
    c) NR

8) Explain how the repertoires in the ELCAR are sequenced.

9) When the ELCAR indicates the possibility of a missing cusp, what type of intervention can be put in place? Give an example of one cusp and the corresponding intervention.

10) The majority of the repertoires in the ELCAR should be assessed using Learn Units (as opposed to unconsequated probe trials). Why is this important?

11) Why is the criterion for mastery of a repertoire in the ELCAR different from the criterion for mastery for an objective that was taught?

12) List the different types of learn units to be used according to students' level of verbal development and briefly explain when to use each type.

13) Explain why it is important to use a variety of stimuli that rotate the irrelevant aspects of the discriminated operant when conducting the ELCAR.

14) Indicate the ELCAR codes (Include the Domain/Section/ Long-term Objectives/Short-term Objective), criteria for mastery, and R+ for the following items:

LTO/STO	ELCAR Code	Criteria	R+
Maintains eye contact for 3 seconds following the command, "Look at me"			
Matches identical pictures (2D) to (3D) objects			
Imitates (copies) lowercase letters a-z			
Emits tacts as introductions			
Uses tissue to wipe nose when needed			
Makes eye contact while answering a question			
Shares toys and books with peers during free play			
Plays independently and appropriately without emitting stereotypy or passivity for 5 minutes in the toy area			

15) Please HAND WRITE the responses below, and indicate how you would record the following, if assessed today.

a) Student met criterion on matching colors

	Long-Term Objectives LTOs	**Crit.**	**Date(s) Identified as Absent**	**Date Identified as Present**
1	Visually matches 5-novel stimuli			
a	Colors	10/10		

b) Student played appropriately for 35% of the five-minute session and emitted stereotypy for the remaining 65%

	Long-Term Objectives LTOs	**Crit.**	**Date(s) Identified as Absent**	**Date Identified as Present**
1	During a 5-minute session in the toy area, plays independently and appropriately without emitting stereotypy or passivity for 90% of the session	3/3		

c) This is a more advanced repertoire of one on which the student did not meet criterion, and therefore was not assessed

	Long-Term Objectives LTOs	**Crit.**	**Date(s) Identified as Absent**	**Date Identified as Present**
1	Match quantity of objects (1-20)			
a	1-10	10/10		

Appendix A: Verbal Behavior Developmental Assessment (VBDA) Cusps for Early Childhood Development

Verbal Foundations	
Cusp:	**Conditioned Reinforcement for 3-Dimensional Stimuli**
Description:	**Student observes 3-D stimuli; 3-D stimuli select out observing responses**
Protocol(s) & References:	**Establish tracking for tabletop stimuli (Du, Broto, & Greer, 2015; Speckman, Longano, & Syed, 2017)**
Abbreviated in VBDA Chart as:	**CR^+ for 3D Stimuli**
Cusp:	**Conditioned Reinforcement for 2-Dimensional Stimuli**
Description:	**Student observes 2-D stimuli (e.g., pictures); 2-D stimuli select out observing responses**
Protocol(s) & References:	**Establish 2-D stimuli and print as reinforcers for visual observing responses (Greer & Han, 2015; Longano, & Greer, 2006; Pereira-Delgado, Greer, Speckman, & Goswami, 2009)**
Abbreviated in VBDA Chart as:	**CR^+ for 2D Stimuli**
Cusp:	**Conditioned Reinforcement for Faces**
Description:	**Student orients or otherwise responds to adult faces; Adult faces select out observing responses**
Protocol(s) & References:	**Establish faces as reinforcing stimuli for observing responses (Maffei, Singer-Dudek, & Keohane, 2014)**
Abbreviated in VBDA Chart as:	**CR^+ for Faces**
Cusp:	**Conditioned Reinforcement for Voices**

Description:	**Student orients or otherwise responds to adult voices; Adult voices select out observing responses**
Protocol(s) & References:	**Establish adult voices as reinforcing stimuli for listening (Greer, Pistoljevic, Cahill, & Du, 2011)**
Abbreviated in VBDA Chart as:	**CR^+ for Voices**
Cusp:	**Conditioned Reinforcement for See-Do Correspondence**
Description:	**Student orients or otherwise responds to adult voices; Adult voices select out observing responses**
Protocol(s) & References:	**Establish reinforcement for see-do correspondence (Du & Greer, 2014; Moreno, 2012)**
Abbreviated in VBDA Chart as:	**CR^+ for See-Do Correspondence**
Cusp:	**Conditioned Reinforcement for the Acoustical Properties of Hear-Say Correspondence without verbal function (Parroting/ Canonical Babbling)**
Description:	**Student emits speech sounds in the absence of a model. Reinforcer is production of the speech sounds themselves**
Protocol(s) & References:	**Stimulus-Stimulus Pairing (Longano & Greer, 2006), Auditory Matching (Choi, Greer, & Keohane, 2015; Speckman-Collins, Park, & Greer, 2007)**
Abbreviated in VBDA Chart as:	**Parroting**

Pre-Joining of the Listener/ Speaker: Listener	
Cusp:	**Reinforcement for responding as a listener (fundamental listener literacy): Responding as a Listener to Speech**

Description:	**Student responds to vocal properties of speech (hear-do)**
Protocol(s) & References:	**Establish phonemic stimulus control for responding to a range of spoken instructions: Listener Emersion (Greer, Chavez-Brown, Nirgudkar, Stolfi, & Rivera-Valdes, 2005)**
Abbreviated in VBDA Chart as:	**Listener Literacy**
Cusp:	**Phonemic Discrimination as a Listener**
Description:	**Student matches sounds, as evidenced by point-to-point correspondence of hear-say responses (echoics)**
Protocol(s) & References:	**Auditory Matching of Single Syllables: Basic (Choi, Greer, & Keohane, 2015; Du, Speckman, Medina, & Cole-Hatchard, 2015; Speckman-Collins, Park, & Greer, 2007)**
Abbreviated in VBDA Chart as:	**Auditory Matching**
Cusp:	**Listening to Story Content Functions as a Reinforcer**
Description:	**Student attends to story and orients toward speaker(s) throughout the duration of the story**
Protocol(s) & References:	**Establish episodic reinforcement story stimuli that keeps changing (Greer & Ross, 2008)**
Abbreviated in VBDA Chart as:	**CR^+ for Listening to Story Content**
Cusp:	**Unidirectional Naming (Conditioned Reinforcement for Names of Stimuli as a Listener)**
Description:	**Student emits untaught listener responses after learning speaker responses or other naming experiences**
Protocol(s) & References:	**Multiple Exemplar Instruction (Fiorile & Greer, 2007; Gilic & Greer, 2009; Greer, Corwin, & Buttigieg, 2011; Greer, Stolfi, Chavez-Brown, & Rivera-Valdez, 2005; Speckman-Collins, Park, & Greer, 2007)**

Abbreviated in VBDA Chart as:	**Unidirectional Naming (UniN)**

Pre-Joining of the Listener/ Speaker: Speaker	
Cusp:	**Conditioned Reinforcement for Correspondence between Hear-Say and Verbal Function (Echoics)**
Description:	**Student emits speech sounds in the absence of a model. Reinforcer is production of the speech sounds themselves**
Protocol(s) & References:	**Echoic-to-Mand Training (Greer & Ross, 2008); Echoic to Tact Training (Greer & Ross, 2008); Rapid Motor Imitation, (Greer & Ross, 2003; Tsiouri & Greer, 2007)**
Abbreviated in VBDA Chart as:	**Echoics**
Cusp:	**Cusp: Correspondence between production (vocal or non-vocal symbol) and delivery of reinforcer (ConTRACT reinforcement)**
Description:	**Student emits mands under relevant motivating conditions**
Protocol(s) & References:	**Echoic-to-Mand Training (Greer & Ross, 2008); Speaker Immersion (Pistoljevic, Cahill, & Casarini, 2010; Ross, Nuzzolo, Stolfi, & Natarelli, 2006)**
Abbreviated in VBDA Chart as:	**Mands**
Cusp:	**Correspondence between production (vocal or non-vocal symbol) and social reinforcement (ConTACT reinforcement)**
Description:	**Student emits tacts as a function of social listener reinforcement**

Protocol(s) & References:	**Echoic to Tact Training (Greer & Ross, 2008), Establish Social Listener Reinforcement through Intensive Tact Instruction or Observational Conditioning (Eby & Greer, 2017; Delgado & Oblak, 2007; Greenberg, Tsang, & Yip, 2014; Lydon, Healy, Leader, & Dolleen-Day, 2008; Pistoljevic & Greer, 2006; Schauffler & Greer, 2006; Schmelzkopf, Greer, Singer-Dudek, & Du, 2017)**
Abbreviated in VBDA Chart as:	**Tacts**
Cusp:	**Transformation of Motivating Conditions Across Mands and Tacts**
Description:	**Student emits mands after learning tacts for the same stimuli and vice versa with no additional instruction**
Protocol(s) & References:	**Establish Transformation of Motivating Conditions Across Mands and Tacts (Nuzzolo-Gomez & Greer, 2004; Singer-Dudek, Park, Lee, & Lo, 2017)**
Abbreviated in VBDA Chart as:	**Transformation of MC across Mands and Tacts**

Joining of the Listener and Speaker **Bidirectional Verbal Operants between people**	
Cusp:	**Conversational Units between an Adult and a Child**
Description:	**Student functions as a listener and a speaker during conversational exchanges with an adult**
Protocol(s) & References:	**Establish Social Listener Reinforcement through Intensive Tact Instruction or Observational Conditioning (Delgado & Oblak, 2007; Pistoljevic & Greer, 2006; Schauffler & Greer, 2006; Schmelzkopf, Greer, Singer-Dudek, & Du, 2017)**

Abbreviated in VBDA Chart as:	**Conv. Units b/w Adult and Child**
Cusp:	**Conversational Units between Two Children**
Description:	**Student functions as a listener and a speaker during conversational exchanges with another child**
Protocol(s) & References:	**Social Listener Reinforcement (Baker, 2014; Reilly Lawson & Walsh, 2007; Sterkin, 2012)**
Abbreviated in VBDA Chart as:	**Conv. Units b/w 2 Children**

Self-Talk: Bidirectional Verbal Operants within the Individual	
Cusp:	**Say-do Correspondence**
Description:	**Student emits relevant behaviors after specifying what they are going to do**
Protocol(s) & References:	**Self-Talk Immersion (Farrell, 2017)**
Abbreviated in VBDA Chart as:	**Say-Do Correspondence**
Cusp:	**Self-Talk Conversational Units in Solitary Play**
Description:	**Student functions his own listener and speaker during solitary play (e.g., imaginative play)**
Protocol(s) & References:	**Self-Talk Immersion (Farrell, 2017)**
Abbreviated in VBDA Chart as:	**Self-Talk Conv. Units in Solitary Play**

Cusp:	Bidirectional Naming (BiN)
Description:	Student emits listener and speaker responses after learning names of stimuli in one modality (listener or speaker) or through incidental naming experiences without direct instruction
Protocol(s) & References:	Multiple Exemplar Instruction (Cao & Greer, 2019; Fiorile & Greer, 2007; Gilic & Greer, 2009; Greer, Corwin, & Buttigieg, 2011; Greer, Stolfi, Chavez-Brown, & Rivera-Valdez, 2005; Greer, Stolfi, & Pistoljevic, 2007; Mosca, 2015), Intensive Tact Instruction (Hotchkiss, 2019; Pistoljevic, 2008)
Abbreviated in VBDA Chart as:	Bidirectional Naming (BiN)
Extensions of Cusp:	**Bidirectional Naming from Fundamental to Multiple Relations**
Description:	Student learns multiple relations (e.g., additional sensory attributes; emotional responses) after learning a single relation
Protocol(s) & References:	Multiple Exemplar Instruction across Senses (Cahill & Greer, 2014; Cao & Greer, 2019; Frias, 2017; Kleinert, 2018; Lo, 2016); MEI across Novel Familiar and Novel Unfamiliar Stimuli (Cao & Greer, 2019; Kleinert, 2019; Lo, 2016); Naming by Exclusion (Greer & Du, 2014)
Abbreviated in VBDA Chart as:	Extensions of BiN

Social Learning: Stimulus Control through Observing Others	
Cusp:	**Observational Performance**
Description:	Student changes behavior to match that of a peer after observing reinforcement (e.g., praise or other generalized reinforcer; access

	to desired item or activity) or punishment (nonspecific correction; disapproval; or natural consequence) delivered to a peer
Protocol(s) & References:	**Peer-Yoked Contingency (Byers, 2016; Gold, 2015); Observational Conditioning-by-Denial Intervention (Lanter, 2018)**
Abbreviated in VBDA Chart as:	**Observational Performance**
Cusp:	**Observational Learning of New Operants**
Description:	**Student acquires new operants as a function of observing others respond and receive reinforcement or corrections**
Protocol(s) & References:	**Peer-Yoked Contingency (Byers, 2016; Gold, 2015); Observational Conditioning-by-Denial Intervention (Lanter, 2018)**
Abbreviated in VBDA Chart as:	**Observational Learning**
Cusp:	**Conditioned Reinforcement-by-Observation**
Description:	**Student learns new reinforcers as a function of observing others receive initially neutral stimuli under conditions of denial.**
Protocol(s) & References:	**Peer-Yoked Contingency (Byers, 2016; Gold, 2015); Observational Conditioning-by-Denial Intervention (Greer & Singer-Dudek, 2008; Lanter, 2018)**
Abbreviated in VBDA Chart as:	**CR^+ by Observation**
Cusp:	**Conditioned Reinforcement for Collaboration**
Description:	**Student's rate of learning should be faster under yoked contingency conditions as compared to individual reinforcement conditions**
Protocol(s) & References:	**Yoked Contingency Game Board (Darcy, 2017)**
Abbreviated in VBDA Chart as:	**CR^+ for Collaboration**

Cusp:	Audience Control
Description:	Student emits behaviors in the presence of one audience but does not emit those behaviors in the presence of a different audience.
Protocol(s) & References:	Social Listener Reinforcement (Reilly Lawson & Walsh, 2007; Sterkin, 2012)
Abbreviated in VBDA Chart as:	Audience Control

Print Joins Prior Cusps

Print Joins Prior Cusps as it relates to reading and writing to learn	
Prior Cusps Join Print:	Conditioned Reinforcement for Observing Book Stimuli
Description:	Student observes 2-D stimuli (e.g., pictures); 2-D stimuli select out observing
Protocol(s) & References:	Establish print and 2D symbols in books functions as reinforcers for observing responses (Buttigieg, 2015; Nuzzolo-Gomez, Leonard, Rivera, & Greer, 2002; Singer-Dudek, Oblak, & Greer, 2011; Tsai & Greer, 2006)
Abbreviated in VBDA Chart as:	CR^+ for Observing Book Stimuli
Prior Cusps Join Print:	Conditioned Reinforcement for Grapheme-Phoneme Correspondence
Description:	Student observes print stimuli and produces correct phoneme without prosthetic reinforcement

Protocol(s) & References:	**Establish correspondence between phonemes and graphemes as reinforcers through learn unit instruction**
Abbreviated in VBDA Chart as:	**CR^+ for Letter Sounds**
Prior Cusps Join Print:	**Conditioned Reinforcement for Grapheme-Phoneme Correspondence for Words and Sentences**
Description:	**Student observes words and sentences and produces phoneme combinations without prosthetic reinforcement**
Protocol(s) & References:	**Establish correspondence between seeing text and speaking words and sentences as reinforcers through learn unit instruction**
Abbreviated in VBDA Chart as:	**Text. Responding**
Prior Cusps Join Print:	**Content of Books Functions as a Reinforcer for Reading**
Description:	**Reads silently with continuous eye tracking for a minimum of 10 minutes**
Protocol(s) & References:	**Establish content of books as reinforcer for reading (Bly, 2019; Cumisky-Moore, 2017; Gentilini & Greer, 2019)**
Abbreviated in VBDA Chart as:	**CR+ for Reading Book Content**
Prior Cusps Join Print:	**Bidirectional Naming Joins Print Stimulus Control in Reading and Writing**
Description:	**Reading results in learning new vocabulary**
Protocol(s) & References:	**Bidirectional Naming from Reading (Helou, 2008)**
Abbreviated in VBDA Chart as:	**Vocab. from Reading**

Prior Cusps Join Print:	**Conditioned Reinforcement for One's Own Writing that Affects the Behavior of a Reader**
Description:	**Writes to affect the behavior of a reader for both technical writing and aesthetic writing**
Protocol(s) & References:	**Writer Immersion (Broto & Greer, 2014; Helou, Lai, & Sterkin, 2007; Madho, 1997; Pellegren, 2015; Reily-Lawson & Greer, 2006; Weber, 2016)**
Abbreviated in VBDA Chart as:	**Functional Writing**
Prior Cusps Join Print:	**Conditioned Reinforcement for Affecting One's Own Writing (Self-Editing)**
Description:	**Student functions his own writer and editor**
Protocol(s) & References:	**Multiple Exemplar Instruction across reader and writer repertoires with a peer-yoked contingency (Pellegren, 2015; Weber, 2016)**
Abbreviated in VBDA Chart as:	**Self-Editing**

Appendix B: Decision Analysis Protocol

CABAS® Decision Protocol
(R. Douglas Greer and Jo Ann Delgado)

Steps Needed to Be in Place in Order to Do a Learn-Unit-in-Context Analysis

Instructions for Supervisors: In order to monitor, teach, and reinforce accurate scientific decisions that ensure pupils receive state-of-the-science instruction, *the professional or trainee must be responsible for all of the instruction associated with the graph check they are doing*. Checks of graphs and feedback to professionals regarding decision points where several individuals have contributed to the decision point can only be attributed to the person in charge of the class, the teacher. Individuals working towards their ranks cannot be indicated as producing a correct or incorrect decision if the above policy is not followed.

Teachers who have mastered the initial graphing components of the Decision Protocol (i.e. 3 data paths ascending, continue) should use the CABAS® DA forms to guide and achieve higher-level decisions. This means, at this point, the teacher has mastered the quiz on graphing decisions and made 20 consecutive correct graphing decisions in the classroom. Refer to the initial contingency shaped and verbally mediated components of the Teacher I Rank. Knowing when to make a decision is a beginning level of competency. The important repertoire involves using the learn unit in context to identify possible sources of learning problems followed by choosing tenable tactics or, where the problem concerns missing verbal cusps, developmental intervention protocols.

Each of the components of the following steps in the Decision Analysis Protocol is followed by a Y (yes) or N (no) response opportunity. When the professional or trainee prepares for a decision, they should follow each of the steps affirming their decision (i.e., the record yes or no and provide rationales if a N is recorded and they defend the decision). *When supervisors or teachers review the decisions and the learn unit context analysis leading to the tactical decision, they too use this checklist as a measure of the decision responses of the trainees or professionals* that they are supervising. Each component is a plus or minus. This measure will lead to a data point for the supervisee's decision accuracy as measured by the supervisor. One measure is the supervisor's count. The other or second use of the measure is the *point-by-point agreement of the supervisor and supervisee relative to each component*. The *percentage of agreement is the measure of the trainees' decision accuracy*. Criterion will be established for each module devoted to the mastery of decisions.

During the first few weeks of trainees' internship or hiring, they should be taught the components of the learn unit in context: (a) Student's potential operant, and (b) Teachers interaction with student that constitutes the learn unit and types of learn units. Subsequently, trainees should learn about (c) setting events, (d) instructional history, (e) phylogenic contributions or environmental trauma. Obviously learning what each of these mean and how to do them is a critical part of what they are to learn.

A. **Graphing Decision**- For this step, the trainee is indicating a decision to stop teaching an objective following the set of rules for making decisions, as described in DTS (2002). This

should be mastered within a week of training. This is as basic as collecting data and it needs to be immediately mastered such that it is second nature (i.e., contingency shaped) within a week of being hired.

(a) Three data paths descending Y/N
(b) Five data paths descending Y/N
(c) Earlier than 3 data points Y/N Rationale:
(d) Criterion achieved for the objective Y/N

B. Curriculum Check- After the lead teacher verifies the decision point above, the lead teacher should indicate that the curriculum is valid for the student. There is no count for this step for the trainee, as the curriculum must be correct in order to proceed forward to identify the source of the instructional problem. The lead teacher signs the form indicating that the curriculum is correct here.

C. Learn Unit Check-

A. Type of Learn Unit: Was the correct type of learn unit delivered? Y/N

Circle if the student has demonstrated the following:

GI &/OR BiN familiar Bin Unfamiliar

If the student has demonstrated Generalized Imitation, then each instructional session should include a demonstration of the correct answers prior to teaching with learn units and indicate the presence of the GI cusp here. If the student has demonstrated bidirectional naming (BiN) with (a) unfamiliar stimuli (First Grade and above), or (b) familiar stimuli (Kindergarten or preschool), two different examples of the response and stimuli are presented prior to the opportunity to respond (Instructional Demonstration Learn Units). Of course, if the student demonstrates any level of BiN, assume GI is present.

If the correct type of learn unit was not in place, implement the correct type of learn unit. If the correct type of learn unit was in place, continue to the next step to ensure that learn units are being delivered accurately.

B. Learn Unit Confirmation. (The learn unit is represented in the BIG EGG illustration of the fields of analysis in the learn-unit-in-context as the egg yolk- see Figure below.) There are two ways to do this and the teacher should conduct at least 1 option (A or B)

- *Option 1*. The first and preferable way is to have Teacher Performance Rate Accuracy (TPRA) observations completed by a qualified supervisor (trained and calibrated to CABAS® standards) showing that learn units have been in place for the target objective. The TPRA should be conducted for the target student and objective in question. Graph this as a probe. Indicate the source of the error(s) on the form (i.e. antecedent, behavior and/or consequence)

  (Y/N) Supervisor doing the TPRA was _____. Date or Dates _____

- *Option 2. Self-report of LU*- To conduct a self-evaluation the teacher should have obtained five consecutive errorless TPRA observations for lessons in the curricular domain (i.e. ELCAR- Academic Literacy, AIL curriculum- Mathematic Standards) for the target decision lesson. They should also have met the calibration standards for doing learn units in the C-PIRK/ELCAR Training Video. Once this calibration has occurred, professionals who are using the ELCAR curriculum mostly may follow and record their responses using the self-report algorithm where they indicate the source of the error(s) in the LU (i.e. antecedent, behavior and/or consequences). This option is used only where the setting does not allow frequent observations (e.g., general education AIL classes, CABAS® self-contained classrooms) of teacher trainees, or TAs. Teachers with CABAS® Master Teacher rank may elect this option. However, they must meet the calibration standards for doing learn units in the C-PIRK/ELCAR Training Video.

Option A & B Learn unit Analysis

- Antecedent Correct: (Y/N) The student was attending when instruction was presented (Potential SD) and unwitting prompts such as looking at the correct answer or providing positional prompts were not present. There was a range of positive exemplars, two negative exemplars for each learn unit, and position and order of positive and negative instances were rotated systematically (i.e. for matching instruction). Materials are unambiguous.
- Behavior Correct: (Y/N) The stimulus or stimuli in the potential SD presentation was/were not faulty. That is, the stimuli were not confusing, jumbled, nor did the stimuli or stimulus have unwitting cues for the correct answers. Behavior was adequately defined so that all professionals can reliably indicate what constitutes a correct and incorrect response. The instructions were presented accurately.
- Reinforcement Correct: (Y/N) Correct answers by the student were followed immediately by delivery of *a known reinforcer* for the particular student or student grouping (i.e., data to affirm the consequence delivered works as a reinforcer for performance and learning for the student or grouping of students). **NOTE**: *Do not assume teacher praise/adult attention functions as a reinforcer*. If praise *alone is used, you must show a functional analysis* or that the child's history of learning affirms that praise alone functions as a reinforcer.
- Corrections Correct: (Y/N) Incorrect answers by the student were followed immediately by corrections that meet the criterion for learn unit corrections. Corrections are key and must be done as follows. The student is given the correct response. In most cases the child will echo or repeat the correct response unless they are providing general education written responses. Next, the learn unit that was missed is presented again ensuring that it is a learn unit. This continues up to three times, if necessary, until the student emits the correct response. The student's corrected response is not reinforced. The teacher affirms the correct response by moving on to the next learn unit in the lesson or session.

If any area of the LU is indicated as no (incorrect), then the decision (context analysis) is not possible so the tactic is to implement the learn unit. Correct the error in your analysis.

If all of the above components were yes, then the appropriate type of learn unit is in place and the curriculum is correct and correctly designed. Given the learn unit prerequisites are in place, a strategic analysis can be done that is a valid learn unit context analysis. The two context categories that are relevant for beginners are: (a) the setting event context surrounding instruction and (b) instructional history where the student is missing prerequisites, missing fluency, or prior instruction interferes. The instructional history source involves changes in what or how the objective is taught. If this is eliminated as the problem, then instructional/experiential history related to verbal developmental cusps become the focus of the analysis. The instructional history analysis is the crux of a 21st century strategic science of teaching that that distinguishes expert professionals from typical practitioners.

Learn Unit Context Analysis.

No decisions are possible unless it can be verified that the type of learn units consistent with the student's verbal behavior developmental status have been used and that the correct curriculum is being taught and the curricular lesson is not faulty.

Step 1. Setting Events Source. In the BIG EGG illustration (see attached) of the learn-unit-in-context, the setting event is represented by the egg white. You must eliminate this as the problem *before moving to the instructional history analysis*. Note that often setting events problems are momentary or limited. For example, overuse of an edible reinforcer may act to interfere with the pupil's rate of learning on a particular day.

Are their events preceding the lesson that may have affected the validity of any part of the learn unit? That is, is there something that happened momentarily, today, or in the even in the last week that affects reinforcement effects of prosthetic reinforcers or natural reinforcers that typically work? Is the child ill, were their conditions in the child's life that may have acted to affect instructional control or the child's community of reinforcers? See DTS for the range of questions and the list of possible tactics to use that are tenable teaching solutions.
Was a setting event involved? Y/N If yes, what was the problem?
Setting event Tactic _____
Setting event decision to stop instruction _____ (e.g., nap, sent home with fever, sent to nurse)

Step 2. Instructional History Source. If setting events are ruled out as the source of the problem then focus on instructional history as the possible source of the problem (i.e., (a) what they have learned from their prior experiences or instruction, something that interferes that they learned in the past or direct instruction and (b) their verbal developmental level). **In the BIG EGG illustration, the instructional history is the egg shell.**

- **Missing Instructional History Source.** Does the student have the prerequisite instructional control repertoires? Does the child respond to vocal instructions (e.g., phonemic awareness, basic listener literacy)? Does the student have the prerequisite objectives in repertoire in order to learn the objective being taught? For example, if you are teaching the child match-to-sample responding, does she know the routine for placing cards that match on top of a match or pointing to a match. Or in another case, the student responds appropriately to some triple digit problems but doesn't meet criterion where the student lacks fluency with

prerequisites such as the "borrowing" function. They lack the prerequisites to learn the objective. Specify whether the student is missing:

- **Prerequisite Skill** (Y/N)- If it is a prerequisite skill that is missing indicate Y and list the skill and next step.
- **Cusp (**Y/N). If it is a cusp that is missing, continue to Step 6

- **Prior Learning Interferes Source:** The student has learned something that interferes with learning the new objective. For example, you are teaching the young child to tact a female deer as a "doe" where the prior learning of dough was learned previously as cookie dough. This effect is referred to as retroactive inhibition. They have learned something that is incompatible with the objective being taught (e.g., teach positive and negative instances of "doe" and "dough" across listener and speaker repertoires such as match, point, tact, intraverbal with visual examples of each. Also, if the student can read, also teach the textual responses). Another example is in math where a child may have been taught the traditional way complete long division (i.e. divide, multiply, subtract, bring down, & repeat) and now is introduced to the partial quotient method and the prior method is incompatible with the new method.

  Prior Learning Interferes Tactic: _____

Step 3. Verbal Behavior Development Source (e.g., verbal behavior development cusps are missing): This is a critical source that determines what kind of instruction can be taught and how the student may be taught. While this expertise is fairly new in the field, all professionals need to master what to look for as soon as possible.

Select one or more of the following that is/are related to the objective being taught. These are listed only as illustrative. The contents of Greer and Ross (2008) and related publications are to be mastered before intervening with developmental protocols.

1) *Does the student lack stimulus control for looking at faces where the faces of others do not reinforce looking at faces or people.* The child does not demonstrate general awareness of others. Does he/she look at new people when they enter the room, does he/she look at you if you talk, does he/she look at other children or does he/she attend only to objects or his/her own stereotypy? Y/N

2) *Is the stimulus control for listening to voices present, even though the content of the words is not present? Voices do not act to reinforce prolonged listening to voices.* This prohibits the learning of phonemic stimulus control. Does he/she listen to you or other adults they read stories? Y/N

3) *Does the student lack conditioned reinforcement for 2D or 3D stimuli,* does he/she lack generalized visual match to sample? That is if you teach him/her to match colors can he/she match shapes without prolonged instruction? Y/N

4) Does the student lack correspondence between seeing and doing or emulating as reinforcing stimuli.? *Does the student lack generalized imitation and conditioned reinforcement for see and do correspondence? That is*, if you show him/her how to do something can he/she imitate what you do, or do you have to do physical prompts? Y/N

5) Is the response of manding or tacting under the control of the relevant reinforcers for emitting echoics? *Does the student echo with point-to-point correspondence* for mand or tact functions (parroting is not verbal)? Y/N

(If the student is preverbal does she demonstrate canonical babbling/parroting where hearing and saying is the reinforcement rather than the function of making the sound to affect another. This irrelevant only of the child is preverbal but vocal)
6) Is the social contractual reinforcement for emitting mands present? *Does the student emit vocal mands*? Y/N
7) *Are the social contact reinforcement stimuli for emitting tacts present?* Y/N
8) *Does the student have transformation of establishing operations across mands and tacts where the relevant reinforcer for each response under the relevant motivational conditions for the reinforcer that determines the response function*?
9) *Doe the student demonstrate bidirectional naming (BiN with familiar stimuli*? Y/N
10) Does the student demonstrate BiN with unfamiliar symbolic stimuli and two and three syllable contrived speech?
1l) *Does the student lack observational learning form observing consequences received by others without direct instruction?*
12) *Does the student lack transformation of stimulus function across saying and writing* that is, if he/she hears a dictated word can he/she say the letters aloud for the word and also write the letters for the word such that if he/she learns one response he/she can do the other without instruction?
13) Is reinforcement stimulus control for looking at books present? This is the prerequisite for teaching reading.
14) Is the reinforcement stimulus control for reading content present? This is critical for reading comprehension.
15) Is reinforcement for emission functional writing present? Part of the functional writing curriculum
16) Is reinforcement for engaging with math stimuli and choosing math in free play present?
17) Does observation of others engaging with stimuli under denial conditions act to condition reinforcers?
18) Is the stimulus control for mutual and combinatorial entailment present? (e.g., frames of coordination, opposition, comparison, distinction, etc.)

Step 4- Phylogeny (Y/N).

Does the student's genetic/physiological make-up interfere with instruction?

If yes, the teacher should consult with the related expert if possible to gather more information and then make changes to the program. Response substitution may be warranted (i.e. point versus vocal response) or rely on successive approximations to the target behavior (vocal approximation versus full word echoic for specific sounds).

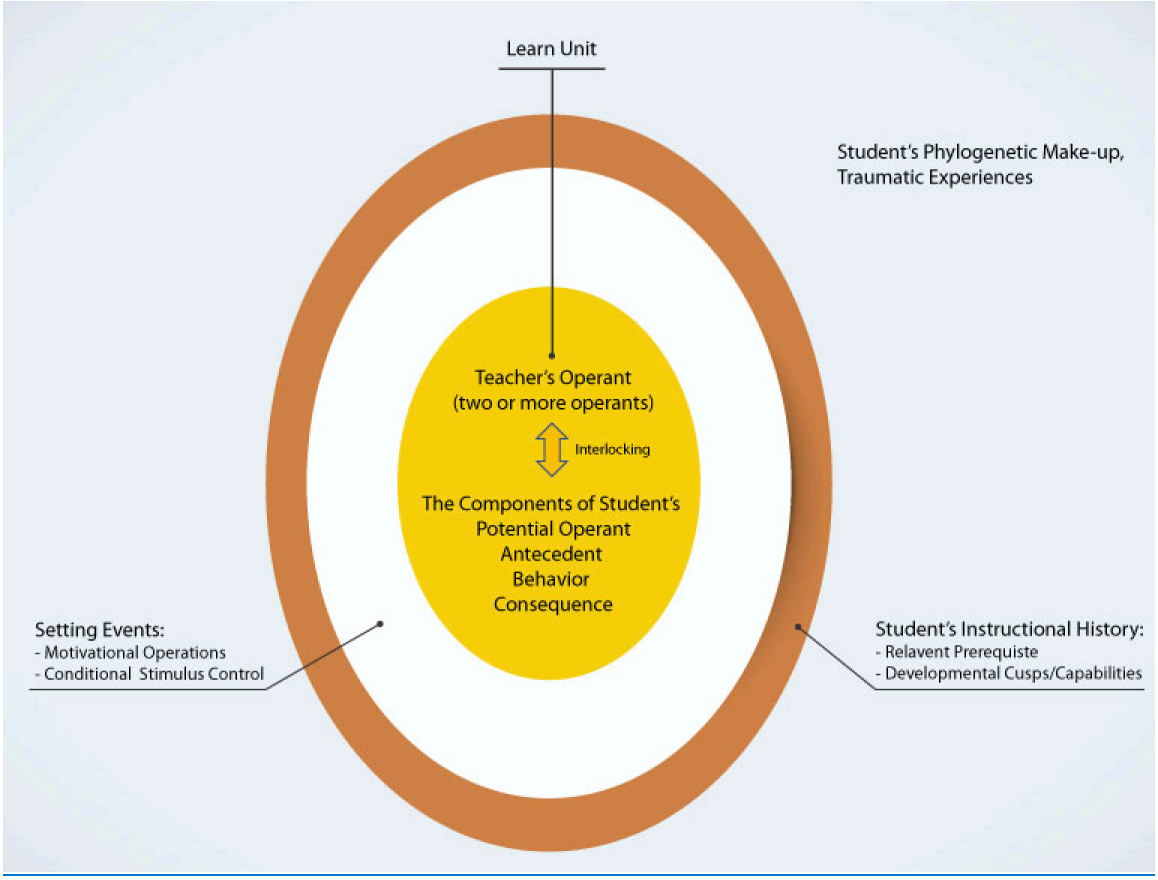
Graphing Teacher Decisions

Teacher I and II Rank Candidates:

Once the form is complete and checked through a supervisor or mentor, graph the results on a bar graph. This should occur after your Supervising teacher signs/initials and provides the results. Each decision (1 completed form) is reflected by one bar on the graph. The bar

represents the total possible steps for that decision (up until where you stopped on the form) and the number of steps correct (point to point correspondence with your supervisor or mentor). Indicate the date and decision level along the x-axis and the number of steps on the y-axis.

Master Teacher Candidates:
Your decision accuracy will be checked through a supervisor on decisions you completed independently. Your supervisor will also check the accuracy of your IOA for trainees' decisions. Graph the percentage of agreement for each decision. This includes point-by-point agreement for each decision component between you and your supervisor. Each decision should be graphed on a line graph, where the x-axis displays the date and the y-axis displays percentage of IOA.



Appendix C: CABAS® Decision Analysis Supervisor Script

***Step 2 Decision Analysis and Beyond**

R. Douglas Greer & Jennifer Weber (October 2019)

Preparation for the Decision Analysis	If Yes….	If No….	Next Steps
Is there a lesson plan or program that incorporates the pre-decision components?	**Continue to next question**	**Prepare a lesson plan**	
Is the objective appropriate for the student?	How do you know that?	*What step do you need to take to assess for this?*	
Is the curricular objective part of the school's curriculum and is it consistent with the relevant educational standards?	What standard is this objective aligned to? (use ELCAR or National/State Standard to determine this)	*What step will you take?*	
Are the materials used to teach the objective unambiguous and clearly related to the educational standard?	*How did you determine this?*	How will you change your materials?	
Was the correct type of learn unit identified in the lesson plan or program? (IDLU, LU, MLU, Probe, Pair Test Trials, Tact presentations, Match-to-Sample while saying the word, etc.)	*How do you know this?* *(ensure that student has appropriate cusp for learn unit presented, BiN, UniN, NinN)*	*How will you change instruction? What type of learn unit will you use?*	
Is there evidence that the correct learn unit was in fact delivered accurately?	How do you know this? (Mentee should either have errorless TPRA or complete self-report LU to show supervisor)	*What kind of training do you need to teach this objective with in-tact learn units?*	
Learn Unit Context Analysis			
SETTING EVENTS ANALYSES			
Are there phylogenetic, anatomical, or physiological differences that suggest the restructure of the instruction or analysis of the source of the problem? (Examples: medical interactions, allergies, psychotropic medications, asthma, immune disorders, migraine headaches, etc.)	**What restructure of this program might be needed to account for this?** **If yes, stop analysis here.*	*Move to next strategic question.*	

Are their conditions in the child's life that may have acted to affect instructional control or the child's community of reinforcers recently?	**What are those conditions? How will you address this?** **If yes, stop analysis here.*	***How do you know?*** **Move onto next question*	
Are there momentary or recent environmental factors interfering with the learn unit presentations? (Examples: Fire drills, a conflict at home or in school, competing classroom contingencies, classroom disruptions)	**How will you address this?** ***What tactic will you use to address this?** **If yes, stop analysis here.*	**Why do you think the problem is not within setting events?** ***If no, move onto next question.**	
INSTRUCTIONAL HISTORY A SOURCES (MISSING PREREQUISITES, RETROACTIVE OR PROACTIVE INHIBITION)			
Does the student have the prerequisite objectives in repertoire in order to learn the objective being taught? **For: The Target Sd** Target topography of behavior Target natural reinforcer Generalized educational reinforcer in repertoire (praise, token, attention)	***How did you determine that?*** **What has the student mastered that ensures you that the student has the prerequisites?**	**What is your next step to address the prerequisite?** **If no, stop analysis here.*	
Has the student previously learned something that is interfering with mastery of this current objective? *Is what is being taught possibly going to interfere with what was taught before?*	**How will you address this? What tactic will you implement?** **If yes, stop analysis here.*	**If no, move onto next strategic question**	

Developmental Instructional History B: Source If the algorithm leads you to determine A POTENTIAL missing STIMULUS CONTROL FOR cusp, follow these questions and steps:		
Strategic Question Related to Possible Missing Cusp	***If Yes….***	***If No….***
Does the student lack stimulus control for looking at faces where the faces of others do not select out looking at faces or people? Does he/she look at new people when they enter the room, does he/she look at you if you talk, does he/she look at other children or does he/she attend only to objects or his/her own stereotypy?	*How will you establish conditioned reinforcement for faces?* *What protocol will you use?*	*How do you know?* **Continue to next question*
Is the stimulus control for listening to voices present, even though the content of the words is not present? Voices do not act to reinforce prolonged listening to voices. This prohibits the learning of phonemic stimulus control. Does he/she listen to you or other adults they read stories?	*How do you know?* **Continue to next question*	*How will you establish conditioned reinforcement for voices?* *What protocol will you use?*
Does the student lack conditioned reinforcement for 2D or 3D stimuli? Does he/she lack generalized visual match to sample?	*How will you establish conditioned reinforcement for 2D or 3D stimuli?* *What protocol will you use?*	*How do you know?* **Continue to next question*
Does the student lack correspondence between seeing and doing or emulating as reinforcing stimuli.? *Does the student lack generalized imitation and conditioned reinforcement for see and do correspondence?*	*How will you establish generalized imitation for this student?* *What protocol will you use?*	*How do you know?* **Continue to next question*
Is the response of manding or tacting under the control of the relevant reinforcers for emitting echoics? *Does the student echo with point-to-point correspondence* for mand or tact functions?	*How do you know?* **Continue to next question*	*How will you establish the student emitting echoics?* *What protocol will you use?*

Is the social contractual reinforcement for emitting mands present? *Does the student emit vocal mands*?	*How do you know?* **Continue to next question*	*How will you establish a mand repertoire for your student?* *What protocol will you use?*
Are the social contact reinforcement stimuli for emitting tacts present?	*How do you know?* **Continue to next question*	*How will you establish a tact repertoire for your student?* *What protocol will you use?*
Does the student have transformation of establishing operations across mands and tacts where the relevant reinforcer for each response under the relevant motivational conditions for the reinforcer that determines the response function?	*How do you know?* **Continue to next question*	*How will you establish transformation of stimulus function across mands and tacts?* *What protocol will you use?*
Doe the student demonstrate bidirectional naming with familiar and unfamiliar stimuli?	*How do you know?* **Continue to next question*	*How will you establish BiN (with familiar and/or unfamiliar stimuli)?* *What intervention will you use?*
Does the student lack observational learning form observing consequences received by others without direct instruction?	*How will you establish Observational Learning?* *Which protocol will you use?*	*How do you know?* **Continue to next question*
Does the student lack transformation of stimulus function across saying and writing? If he/she hears a dictated word can he/she say the letters aloud for the word and also write the letters for the word such that if he/she learns one response he/she can do the other without instruction?	*How will you establish transformation of stimulus function across saying and writing?* *What protocol will you use?*	*How do you know?* **Continue to next question*
Is reinforcement stimulus control for looking at books present?	*How do you know?* **Continue to next question*	*How will you establish conditioned reinforcement for looking at books?* *What protocol will you use?*
Is the reinforcement stimulus control for reading content present?	*How do you know?* **Continue to next question*	*How will you establish conditioned reinforcement for content of reading?* *What procedure will you use?*

Is reinforcement for emission functional writing present?	*How do you know?* **Continue to the next question*	*How will you establish functional writing for your student?* *What procedure will you use?*
Is reinforcement for engaging with math stimuli and choosing math in free play present?	*How do you know?* **Continue to the next question*	*How will you establish conditioned reinforcement for math?* *What procedure will you use?*
Does observation of others engaging with stimuli under denial conditions act to condition reinforcers?	*How do you know?* **Continue to the next question*	*How will you establish this?* *What procedure will you use?*
Is the stimulus control for mutual and combinatorial entailment present? (e.g., frames of coordination, opposition, comparison, distinction, etc.)	*How do you know?*	*How will you establish this stimulus control?*
Next Step for Student:		

Questions Completed: _____
Correct Responses: _____
Percentage Correct: _____

Fred S. Keller School

The Fred S. Keller School (FSK), named for the behavior analyst and pioneer 'in experimental psychology, Fred Simmons Keller, has campuses in Yonkers (Westchester County) and Palisades (Rockland County), New York. FSK is an internationally recognized behavior analytic preschool and early intervention program for children from eighteen months through age five with and without disabilities. The school serves as a research and demonstration center for state of the science differentiated instruction and curriculum-based assessment. Research reports children in the CABAS® systems learn 4 to 7 times more than control or baseline educational practices with which it has been compared. See references below for independent evaluations.

The school is one of several Comprehensive Application of Behavior Analysis to Schooling or CABAS® (Selinske, Greer, '& Lodhi, 1991) accredited programs in the United States and Surry, England. CABAS® programs are characterized by the following components of quality: individualized instruction, continuous measurement of teaching and student responses or curriculum based assessment, graphic display of student learning and achievement of educational standards, the use of scientifically-tested tactics for instructional decision making, logically and empirically tested curricula and curricular sequences, educationally and socially significant goals of instruction, positive teaching environments where coercive procedures are avoided, and teachers who are strategic scientists of pedagogy (Greer, Keohane, & Healy, 2002). CABAS® applies the underlying principles of behavior analysis and advanced principles of teaching and verbal development to all components of education which include the role of the students, parents, teachers, school supervisors and administrators, and its board and university training program (Greer, 1994).

The Fred S. Keller School offers three different program options for children early intervention services, a full and half-day preschool program and our newly expanded integrated preschool program.

Early intervention is a public program funded by New York State and Westchester County governments with no cost to parents. Eligibility for early intervention services is determined by the county prior acceptance into our school program. Early intervention services provide family training and individualized instruction to toddlers with educational disabilities sixteen months to three years of age. Most students are previously identified as having one or more of the following disabilities: speech and language delays, social skills deficits, and/or characteristics of autism and pervasive developmental delays. Services are delivered in both the home and school setting by highly trained CABAS® board certified behavior analysts who are licensed special education and general education teachers. At the onset of services, supervision services are intensive as the coordinator accompanies the teachers to aide in the assessment and creation of the student's program portfolio. Weekly meetings are held and attended by all members of the educational team to review individual student performance. Changes in students' programs are continuously made as needed and always data driven.

The Fred S. Keller School provides a continuum of behavior analytic educational services for those students who continue to need special education after early intervention. The full-day and half-day preschool classrooms serve educationally disabled children from the ages of 3-5. We

provide classrooms with either a 6:1:2 or 6:1:1 ratio contingent upon the student's level of verbal behavior. Once our students acquire the necessary academic and self-management repertoires needed to enter a typical kindergarten, an integrated preschool program is offered. This classroom provides opportunities for these children with disabilities to be educated with their non-disabled peers. Seventy-five percent of these children do graduate into typical kindergarten classrooms with and without support.

We also accept a limited number of typically developing children on a tuition basis. These children are assured of entering kindergarten performing at first grade level in most subject areas. Some scholarships are available for these children.

For more information visit https://www.cabasschools.org/fred-s-keller-school or email jspeckman@fredskeller.com

FASST
Foundation for the Advancement of a Strategic Science of Teaching

FASST is a non-profit corporation and shall operate exclusively for educational and charitable purposes. The purpose of FASST is to support, enhance, develop, and expand the strategic science of teaching and other behavior analytic initiatives committed to a science of teaching for all learners through research, application, dissemination, and replication, of the science of behavior analysis.

FASST's mission is to contribute to the well-being of society by promoting teaching, professional education, collaboration, research, and competency driven applications of the science of behavior analysis to solve problems in the home, school, community, and workplace through application in settings that include public, charter and private schools; clinics; learning centers; and in the context of tutoring. FASST strives to create, promote, and otherwise enhance domestic and global dissemination of scientifically sound and otherwise reliable information on behavioral services and science, and to promote collaboration that advances this goal. Core objectives of FASST are to promote collaboration among those engaged in behavior analysis, to support development and expansion of a learner- driven science of teaching for all children, to improve ongoing and future applications of behavior analysis directly and through the education of those providing and receiving behavior analytic services, and to raise awareness of and otherwise link the public and behavior analyst with evidence-based and otherwise scientifically sound behavior analytic resources.

For more information, visit www.scienceofteaching.org or e-mail to fasst2018@gmail.com